

SUMMARY TABLE: MATHEMATICS GRADE 10

Sub-Strand 3.4: Linear Motion — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 3.4: Linear Motion
Driving Question	DRIVING QUESTION: How can we use distance, time, speed, velocity and graphs to describe and predict how one car overtakes another on a Kenyan road?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Phenomenon: Overtaking on the Naivasha Highway	What did we observe in the clips? Describe in your own words what car A (blue Toyota Axio), car B (white Subaru) and the oncoming lorry L were doing, and what happened when B returned to its lane.	What did we notice that puzzled us? Record how the speedometers looked, how car A looked from car B, and what questions we now want to answer about how long, how far and how fast.	What is our first explanation? Write your best current idea for why car A looks slow from car B even though both speedometers are high, and why the oncoming lorry matters, and say what you still need to find out.
Lesson 2 Distance, Displacement, Speed, Velocity and Acceleration: Describing Motion on Kenyan Roads	Juma's boda boda trip: 4 km distance but 2 km displacement east of the stage. Naivasha Express matatu: 90 km in 1.5 h. Cyclist Atieno: 12 km distance but displacement 0. In the overtaking clip, car A stays steady at 72 km/h, car B speeds up from 72 km/h to 108 km/h, and oncoming lorry L travels the opposite way.	Distance is total path length (scalar); displacement is straight-line change in position with direction (vector). Speed uses distance; velocity uses displacement and direction. Acceleration is the rate of change of velocity in m/s^2 : positive when speeding up, zero for constant velocity, negative (deceleration) when slowing down.	Car A has constant velocity so zero acceleration. Car B has positive acceleration because its velocity increases. Lorry L has constant negative velocity because it travels towards Naivasha, opposite to the positive direction towards Nakuru, and zero acceleration. A speedometer shows speed only, so velocity is needed to describe direction as well.
Lesson 3 Calculating Velocity and Acceleration: How Fast, and How Quickly Does the Speed Change?	We observed that Car A covers 150 m in 7.5 s, Car B's speedometer rises from 72 km/h to 108 km/h in 5 s and that the oncoming lorry L moves towards Naivasha. We observed that the matatu slows from 20 m/s to 0 in 10 s.	We learned that velocity = displacement/time and $a = (v - u)/t$, that km/h is converted to m/s by dividing by 3.6, and that a negative sign shows direction (the lorry, -20 m/s) or deceleration (the braking matatu, $a = -2 \text{ m/s}^2$).	We can explain that A has velocity +20 m/s with zero acceleration, B has acceleration 2 m/s^2 while it speeds up from 20 to 30 m/s, and L has velocity -20 m/s because it travels in the opposite direction, so the numbers describe how each vehicle moves in the overtaking.
Lesson 4	We observed data for a parked matatu 5 km	We learned to choose scales, label axes	We can draw a graph of displacement

Drawing Displacement-Time Graphs: Parked, Steady and Returning Movements	from Naivasha town, a steady car at 60 km/h with points (0,0), (30,30), (60,60), and cyclist Atieno with points (0,0), (30,6), (40,6), (60,0) in (min, km). We saw that the displacement stays the same, rises steadily, or returns to zero.	with units, plot points and join them with a ruler to draw displacement-time graphs. A parked vehicle gives a horizontal line, a steady car gives a straight rising line, and a returning cyclist gives a line that rises, goes flat and then falls to zero.	against time for a stationary, steady or returning movement and describe its shape. This helps us describe the steady motion of Car A in the overtaking, and shows we need more tools, such as reading steepness and velocity-time graphs, to describe Car B speeding up.
Lesson 5 Interpreting Displacement-Time Graphs: Gradient, Rest and Meeting Points	We observed that the graph of Atieno's ride has three sections with gradients 12 km/h, 0 and -18 km/h, and that the lines for the matatu leaving Naivasha at 60 km/h and the car leaving Nakuru at 90 km/h cross at 24 min and 24 km from Naivasha.	We learned that the gradient of a displacement-time graph is velocity, that its sign shows direction (negative means the opposite way, like the oncoming lorry L), that a flat line means rest, and that the intersection of two graphs shows where and when two bodies meet.	We can explain that the vehicles meet when their graphs cross because they are at the same displacement at the same time. This helps us describe the motion of Car A, Car B and the oncoming lorry L, but we still need velocity-time graphs and acceleration to explain how Car B speeds up when overtaking.
Lesson 6 Drawing Velocity-Time Graphs: Speeding Up, Steady Speed and Slowing Down	What did we observe? Example: Car A's velocity-time graph is a horizontal line at 20 m/s. Car B's graph rises in a straight line from 20 m/s at $t = 0$ to 30 m/s at $t = 5$ s and then is horizontal until $t = 7.5$ s. The braking matatu's graph falls in a straight line from 20 m/s at $t = 0$ to 0 m/s at $t = 10$ s.	What did we learn? Example: A rising line shows speeding up (positive acceleration, e.g. 2 m/s^2), a horizontal line shows steady speed (zero acceleration), and a falling line shows slowing down (deceleration, e.g. $a = -2 \text{ m/s}^2$). A velocity-time graph has t (s) on the x-axis and v (m/s) on the y-axis.	How does this help us explain the overtake? Example: The graphs show that Wanjiru in Car B speeds up gradually from 72 km/h to 108 km/h in 5 s and then holds that speed, while Mr Kamau in Car A stays at 72 km/h. Next we need to find how far each car travels and how their speeds compare.
Lesson 7 Interpreting Velocity-Time Graphs: Gradient for Acceleration, Area for Distance	We observed that Car B's velocity-time line slopes upward from 20 m/s to 30 m/s in 5 s, then stays flat to 7.5 s; Car A's line is flat at 20 m/s; the braking matatu's line slopes downward from 20 m/s to 0 in 10 s. The areas under the graphs matched the split-time table.	We learned that the gradient of a velocity-time graph gives acceleration (B: 2 m/s^2 ; matatu: -2 m/s^2 , deceleration 2 m/s^2) and the area under the graph gives distance travelled (B: $125 + 75 = 200 \text{ m}$; A: 150 m ; matatu: 100 m).	We can explain that Car B travels 50 m more than Car A during the overtake because it sped up at 2 m/s^2 , and that velocity-time graphs let us describe and predict the motion of the cars on the Naivasha highway using gradient and area.
Lesson 8 Relative Speed: Why Does the Overtaken Car Look Slow?	We observed that at $t = 0$ both car A and car B show 72 km/h (20 m/s), yet B gains 1, 4, 9, 16, 25, 36, 49 and 64 m on A at $t = 1, 2, 3, 4, 5, 6, 7$ and 7.5 s, and oncoming lorry L moves towards them at 20 m/s.	We learned that relative speed is found by subtracting speeds for bodies moving in the same direction (B relative to A = $30 - 20 = 10 \text{ m/s} = 36 \text{ km/h}$, growing from 0 at $t = 0$ to 10 m/s at $t = 5$ s) and by adding speeds for bodies moving in opposite directions (B and L: $30 + 20 = 50 \text{ m/s} = 180 \text{ km/h}$), always stating the viewpoint. The area under B's relative velocity graph, $25 + 25 = 50 \text{ m}$, gives the extra distance B gains.	Car A looks slow from car B because they move in the same direction, so their relative speed is only 10 m/s (36 km/h) and was 0 at the start, even though both speedometers read high. The oncoming lorry looks very fast from B because it closes at 50 m/s (180 km/h), which is why the overtake needs careful checking in the next lesson.
Lesson 9 Applying Linear Motion: Is Wanjiru's Overtake	What we observed: In the Naivasha clip, overtaking car B took 7.5 s to pass overtaken car A. In that time B travelled 200	What we learned: We can combine acceleration ($a = 2 \text{ m/s}^2$), areas under velocity-time graphs and relative speed to	How this explains the phenomenon: With the lorry 500 m away, the margin is 150 m, or 3.0 s, so Wanjiru's overtake is safe. If the

Safe?	m, A travelled 150 m and oncoming lorry L travelled 150 m. B gained 50 m on A.	decide if an overtake is safe. B and L together use 350 m of road. The safety margin is the distance to the lorry minus 350 m, and dividing by the closing speed of 50 m/s gives the margin in seconds.	lorry were only 400 m away, the margin would be 50 m, or 1.0 s, which is unsafe. Both cars can have high speedometer readings, but safety depends on relative speeds and the road available, not on speedometer readings alone.
Lesson 10 Final Model and Answer to the Driving Question: Overtaking on the Naivasha Highway	We observed Car B (white Subaru) overtake Car A (blue Toyota Axio) near Naivasha: both start at 72 km/h (20 m/s), B speeds up to 108 km/h (30 m/s) in 5 s and is back in lane at 7.5 s, while the oncoming lorry L approaches at 20 m/s.	We learned that velocity = displacement/time and $a = (v - u)/t$; that the gradient of a velocity-time graph gives acceleration (2 m/s^2) and the area gives distance (B 200 m, A 150 m); and that relative speed is the difference of speeds in the same direction (10 m/s) and the sum in opposite directions (50 m/s).	We explained that A looks slow from B because B relative to A is only 10 m/s (36 km/h), and that the overtake is safe because B and L need 350 m of road, leaving 150 m (3.0 s) of the 500 m; with the lorry at 400 m the margin is only 1.0 s, which is unsafe.

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